

Lipid Regulating Drugs

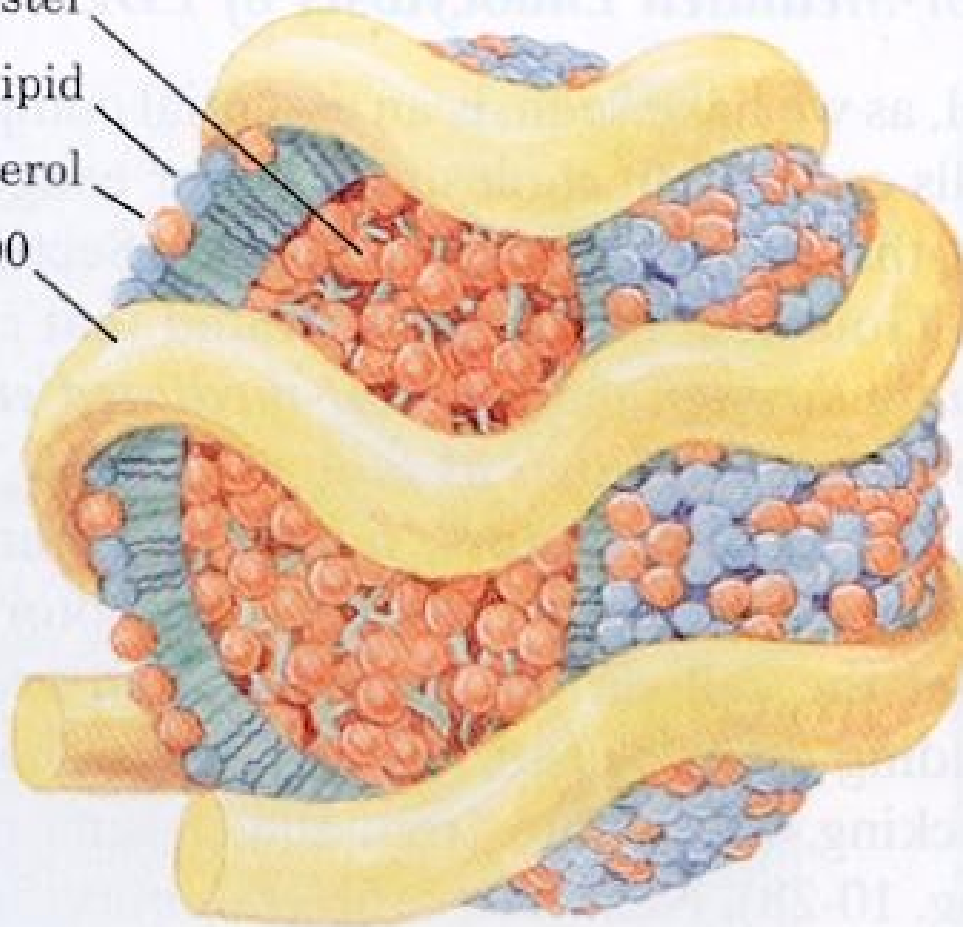
Introduction

- ❖ Plasma lipids are **transported** in complexes called lipoproteins.
- ❖ Metabolic disorders that involve elevations in any lipoprotein species are termed **hyperlipoproteinemias** or **hyperlipidemias**.
- ❖ **Hyperlipidemia** denotes increased levels of triglycerides.
- ❖ The two major clinical sequelae of hyperlipidemias are **acute pancreatitis** and **atherosclerosis**.

Structure of lipoproteins

- ❖ Lipoproteins have hydrophobic core regions containing cholesteryl esters and triglycerides
 - surrounded by **unesterified cholesterol, phospholipids, and apoproteins**.
- ❖ The protein components, known as apolipoproteins or apoproteins,
 - provide **structural stability to the lipoproteins** and also may function as ligands in lipoprotein–receptor interactions or as **cofactors** in enzymatic processes that regulate lipoprotein metabolism.
- ❖ In all spherical lipoproteins, the most **water-insoluble lipids** (cholesteryl esters and triglycerides) are core components, and the more polar, **water-soluble components** (apoproteins, phospholipids, and unesterified cholesterol) are located on the surface.

Cholesteryl ester
Phospholipid
Unesterified cholesterol
Apolipoprotein B-100



LIPOPROTEIN DISORDERS

- Lipoprotein disorders are detected by measuring lipids in serum after a 10-hour fast.
- Risk of heart disease increases with concentrations of the atherogenic lipoproteins, is inversely related to levels of HDL.

LDL cholesterol	
Optimal	< 100
Near optimal/borderline	100–129
Borderline high	130–159
High	160–189
Very high	≥ 190
Total cholesterol	
Desirable	≤ 200
Borderline high	200–239
High	≥ 240
HDL cholesterol	
Low	≤ 40
High	≥ 60

Simple Classification of Hyperlipidemias

	TC	TG	HDL
Hypercholestrolemia	↑		
<i>Hypertriglyceridemia</i>		↑	
<i>Mix Hyperlipidemia</i>	↑	↑	
hypoalphalipoproteinemia			↓

Agents managing hyperlipidemia

-----**Drugs lowering TC** (total cholesterol)

-----**Drugs lowering TG**

-----**Drugs increasing HDL**

Classes of Drugs

I-Agents targeting endogenous cholesterol:

a-Statins.

b-Fibrates.

C-Nicotinic acid.

II-Agents Targeting Exogenous Cholesterol

a-Cholesterol Uptake Inhibitors,

e.g. ezetemibe.

b- Bile acid binding resins

e.g. colestipol & cholestyramine

Competitive inhibitors of HMG-CoA reductase (statins)

- ❖ These compounds are structural analogs of HMG-CoA (3-hydroxy-3-methylglutaryl-coenzyme A) .
- ❖ **Drugs**.....Lovastatin, atorvastatin, fluvastatin, pravastatin, simvastatin, rosuvastatin, and pitavastatin belong to this class.
- ❖ They are most effective in reducing **LDL**.
- ❖ Other effects include **decreased oxidative stress** and **vascular inflammation** with increased **stability of atherosclerotic lesions**.

ADME

Absorption

- ❖ After oral administration, intestinal absorption of the statins is variable (30%–85%)
- ❖ Absorption generally (with the exception of **pravastatin** and **pitavastatin**) is **enhanced by food**
- ❖ **Forms.....**All the statins, except **simvastatin** and **lovastatin**, are administered in the **β-hydroxy acid form**,
 - ✓ Simvastatin and lovastatin are administered as inactive lactones
 - ✓ that must be transformed in the liver to their respective β-hydroxy acids,
 - Simvastatin.....> simvastatin acid, and
 - Lovastatin.....> lovastatin acid.

ADME

- ❖ metabolites have some activity ...except **fluvastatin** and **pravastatin**,
- ❖ more than 95% of statins and their metabolites are protein bound,
- ❖ pravastatin and its metabolites....only 50% bound
- ❖ Peak plasma concentrations1–4 h.
- ❖ $t_{1/2}$ of the parent compounds 1–4 h,
 - except in the case of atorvastatin and rosuvastatin.....**about 20 h**
 - The may contribute to their greater cholesterol-lowering efficacy
 - **Simvastatin**.....12 hr

ADME...

Metabolism

- ❖ All statins would have first-pass hepatic uptake of all statins, mediated primarily by the organic anion transporter **OATP1B1**
- ❖ **Hepatic cholesterol synthesis is maximal between midnight and 2:00 am**
- ❖ Statins with $t_{1/2}$ of 4 h or less should be taken the evening
 - ✓ **Atorvastatin** and **rosuvastatin** both have longer half-lives and
 - may be taken at other times of day to optimize adherence

Mechanisms of action

- ❖ Statins exert their major effect—reduction of LDL levels—
- ❖ through a competitively inhibiting HMG-CoA reductase.
- ✓ reduce the conversion of HMG-CoA to mevalonate,
 - an early and rate-limiting step in cholesterol biosynthesis.
- ❖ They affect Statins blood cholesterol levels by
- ❖ inhibiting hepatic **cholesterol synthesis**,
 - which results in increased expression of the LDL receptor gene

Mechanisms of action..

- ❖ They can also reduce **LDL** levels by
 - ✓ enhancing the removal of LDL precursors (VLDL and IDL) and
 - ✓ decreasing hepatic production.
- ❖ The reduction VLDL in hepatic **VLDL** production induced by statins is thought to be mediated by reduced synthesis of **cholesterol**, a required component of VLDLs
- ❖ Maximal effects on plasma cholesterol levels are achieved within **7–10** days
- ❖ Effective in all cases caused by an increase LDL-C except *hoFH*

Toxicity and Drug interaction

❖ Hepatotoxicity.....Rare case (1 case per million)

- **Liver enzymes** should be evaluated in patients with clinical symptoms suggestive of liver injury

Myopathy

❖ **It is the major adverse effect associated with statin use**

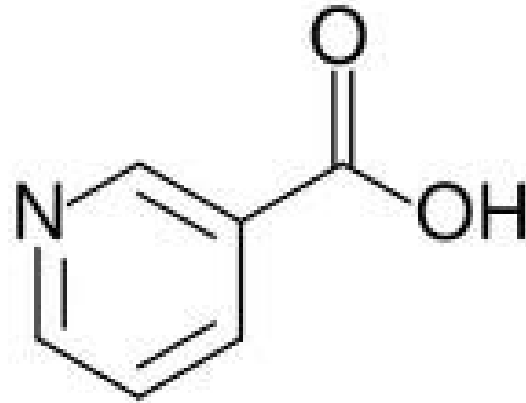
- ✓ It refers to a broad spectrum of muscle complaints, ranging from mild **muscle soreness** or **weakness (myalgia)** to life-threatening **rhabdomyolysis**

- ❖ The risk of muscle adverse effects increases by factors inhibiting statin catabolism
 - ✓ including advanced age (especially > 80 years of age),
 - ✓ hepatic or renal dysfunction, perioperative
 - ✓ untreated hypothyroidism
- ❖ Measurements of creatinine **kinase are not routinely necessary unless** the patient also is taking a drug that enhances the risk of myopathy
 - ✓ Necessary e.g when it is given with Gemfibrozil,
- ✓ **Rosuvastatin** and **pravastatin** may be better tolerated than other statins
 - should be considered in patients with a history of **myalgias** with **other statins**.

Contraindications

- Pregnancy category X
- Not given during lactation
- Active liver diseases

NICOTINIC ACID (NIACIN)

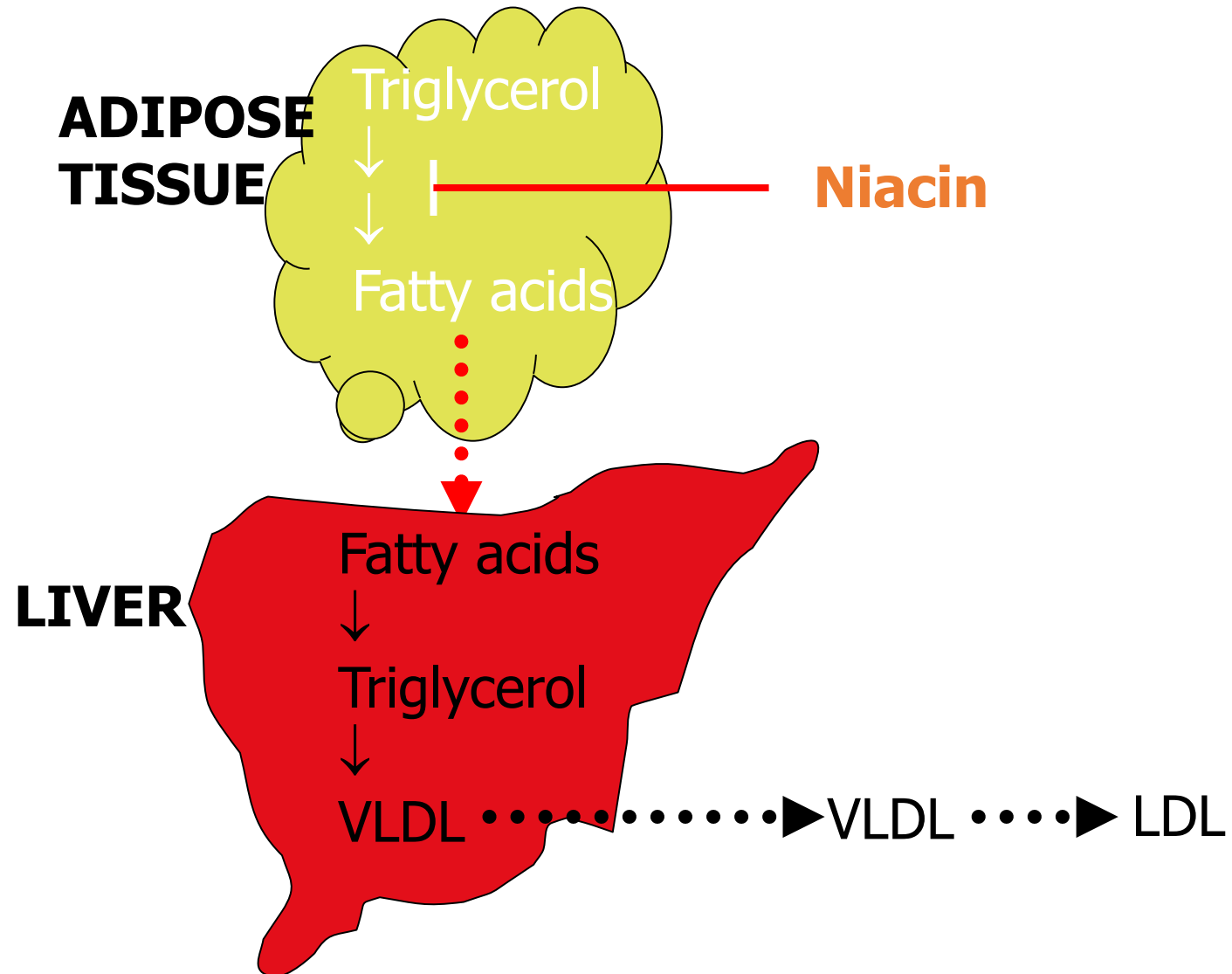


Nicotinic Acid (Niacin, Vitamin B₃)

Mechanisms

1. **In adipose tissue:** it binds to adipose **nicotinic acid receptors**, this will lead to decrease in **free fatty acids** mobilization from adipocytes to the liver resulting in **↓ TG** and thus **VLDL synthesis**.
2. **In liver:** niacin inhibits **hepatocyte diacylglycerol acyltransferase-2**, a key enzyme for TG synthesis. Thus, it decreases VLDL production (decreased TG synthesis and estrification).
3. **In plasma:** it **increases LPL activity** that increases clearance of VLDL & chylomicron.
4. May **↓ HDL catabolism**

Lipolysis inhibition by Niacin



Nicotinic Acid (Niacin, Vitamin B₃)

Efficacy:

TC ↓ 25%

LDL ↓ 10-25%

HDL ↑ 10-40%

TG ↓ 20-50%

Indications:

High LDL (and/or VLDL)

Combined hyperlipidemia (including low levels of HDL--, approved for elevating HDL levels)

Niacin – Adverse effects

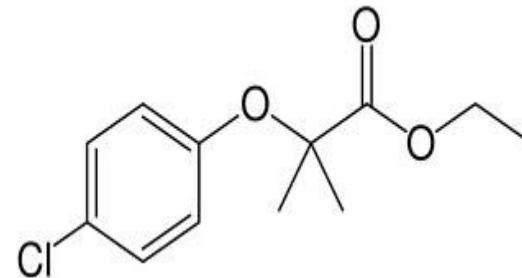
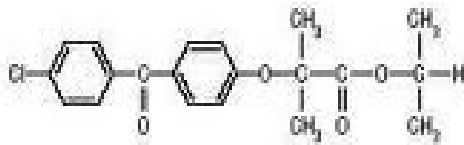
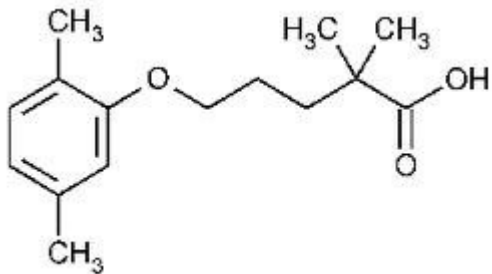
- Flushing
 - Harmless **cutaneous vasodilation**, VERY uncomfortable
 - Occurs after drug is started or ↑ dose
 - Lasts for the first several weeks
 - Relieved by giving aspirin 30 minutes before dosing
- Pruritis, rashes, dry skin
- Nausea and abdominal discomfort

Niacin – Adverse effects

- Hepatotoxicity
 - Rare true hepatotoxicity occurs
 - Monitor liver functions regularly
 - Liver injury is less likely with Niaspan
- Hyperuricemia
 - Occurs in about 1/5 of pts
 - Occasionally precipitates gout
- Carbohydrate tolerance may be moderately impaired (hyperglycemia)
 - Reversible
 - Can be given to diabetics receiving insulin

FIBRIC ACID DERIVATIVES (FIBRATES)

Gemfibrozil, fenofibrate and clofibrate



Mechanism of Action

- ❖ The **mechanisms** by which fibrates lower lipoprotein levels, or raise HDL levels, remain unclear.
- ❖ Many of the effects of **these compounds** on lipids are **mediated by their interaction with PPARs**.
 - PPARs regulate gene Transcription.
- ❖ **binding to PPAR α Fibrates**
 - ✓ reduce triglycerides through PPAR α -mediated stimulation of fatty acid oxidation,
 - ✓ increased LPL synthesis, and
 - ✓ reduced expression of apo C-III

Mechanism of Action

- ❖ Reduced hepatic production of apo C-III,
- ❖ Apo C-III serve as an **inhibitor of lipolysis** and **receptor-mediated clearance**, would enhance the clearance of VLDLs
- ❖ Fibrate-mediated increases in HDL-C are due to PPAR α stimulation of **apo A-I** and **apo A-II** expression (effectiveness, Fenofibrate >> gemfibrozil)
- ❖ Most fibrates have potential antithrombotic effects

Effects on Lipoprotein Levels

- ❖ Effects of these agents on lipoprotein levels differ widely, depending on
 - ✓ the starting lipoprotein profile,
 - ✓ the presence or absence of a genetic hyperlipoproteinemia,
 - ✓ the associated environmental influences, and
 - ✓ the specific fibrate used.
- ✓ Patients with type III hyperlipoproteinemia (dysbetalipoproteinemia)
 - sensitive responders to fibrates
- ❖ Elevated **triglyceride** and **cholesterol** levels are dramatically lowered
- ❖ Patients with the following condition respond to fibrates
 - tuberous xanthomas
 - palmar xanthomas
 - intermittent claudication
 - Angina

Effects on Lipoprotein Level...

❖ In patients with mild hypertriglyceridemia (e.g., triglycerides < 400 mg/dL),

- ✓ decreases triglyceride levels by up to 50% and
- ✓ increases HDL-C concentrations by about 15%;
- ✓ LDL-C levels may be unchanged or increase.

❖ drugs of choice for treating severe **hypertriglyceridemia** and the **chylomicronemia syndrome**.

- ✓ increasing TG clearance and decreasing hepatic TG synthesis

❖ **fibrate maintenance therapy + a low-fat**

- ✓ keep triglyceride levels well below **1000 mg/dl**...prevent episodes of pancreatitis

ADME

- ❖ Absorptionabsorbed rapidly and efficiently (>90%) when given with a meal
- ❖ Peak plasma concentrations.....within 1–4 h
- ❖ Binding..... > 95% of these drugs in plasma are bound to protein, nearly **exclusively to albumin**
- ❖ $t_{1/2}$
 - ✓ range from 1.1 (gemfibrozil) to 20 h (fenofibrate)
- ❖ **Distribution**
 - ✓ widely distributed throughout the body
 - ✓ **concentrations in liver, kidney, and intestine exceed the plasma level.**
- ❖ **Excretion**
 - ✓ predominantly as glucuronide conjugates (60%–90%) in the urine
 - ✓ smaller amounts appearing in the feces.

Adverse Effects and Drug Interactions

- ❖ S/E....usually are well tolerated.
- ❖ GI side effects occur in 5% of patients.
- ❖ **Infrequent side effects**
 - ✓ rash, urticaria, hair Loss, myalgias, fatigue, headache, impotence, and anemia.
- ❖ Minor increases in liver transaminases and ALP have been reported

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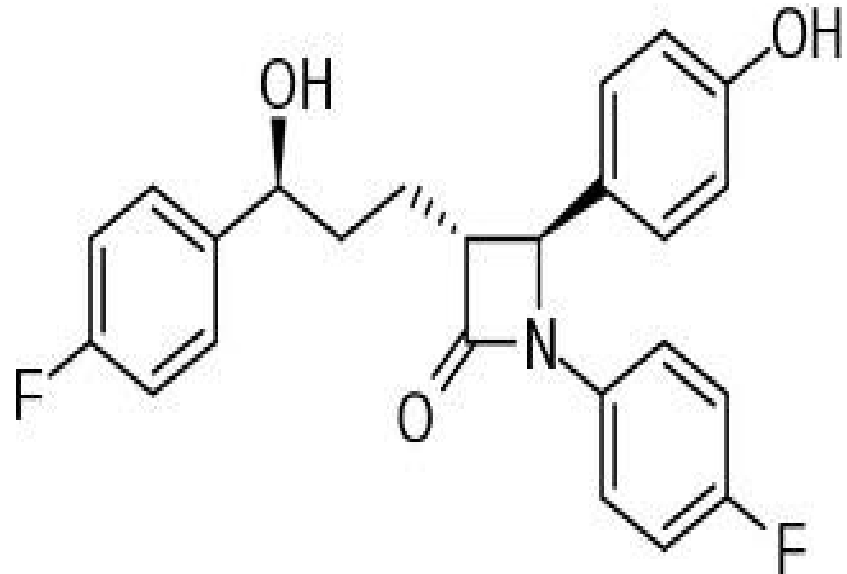
Adverse Effects and Drug Interactions...

- ❖ All of the fibrates increase the lithogenicity of bile.
- ❖ Clofibrate use has been associated with increased risk of gallstone formation.
- ❖ **C/I....**Fibrates should not be used by children or pregnant women.

- ✓ A myopathy syndrome.....In subjects taking clofibrate, gemfibrozil, or fenofibrate
- ✓ gemfibrozil + higher doses of statins....5 %
 - Gemfibrozil inhibits hepatic uptake of statins by OATP1B1 and competes for the same glucuronosyl transferases that metabolize most statins
- ✓ Fbrates+ rosuvastatin low doses (5–10 mg).....Myopathy syndrome is high
- ✓ **Fenofibrate** is glucuronidated by enzymes that are not involved in statin glucuronidation
 - fenofibrate-statin combinations are less likely to cause myopathy

Cholesterol absorption inhibitors

Ezetimibe



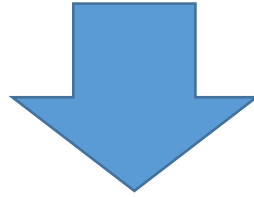
- ❖ Ezetimibe is the first compound approved for lowering **total** and LDL-C (reduce by 20 %). used as adjunctive therapy with statins.
- ❖ It inhibits cholesterol absorption by enterocytes in the small intestine.

Mechanisms of Action

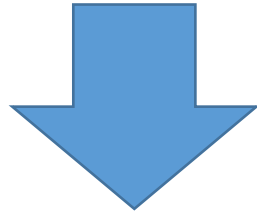
- ❖ Ezetimibe inhibits **luminal cholesterol** uptake by **jejunal enterocytes**, by **inhibiting the transport protein NPC1L1**
- ❖ Reduce cholesterol absorption by 54 %
 - **precipitating a compensatory increase in cholesterol synthesis**
 - **Can be reduced by statins**

Ezetimibe...

❖ reduction in the incorporation of cholesterol into chylomicrons



❖ diminishes the delivery of **cholesterol** to the liver
by **chylomicron remnants**.



❖ stimulates expression of the hepatic genes regulating LDL receptor expression

✓ decrease atherogenesis directly.....chylomicron remnants highly atherogenic **LPs**

✓ Ezetimibe reduces **LDL-C levels by 15%–20%**

ADME

❖ **Highly bioavailable**

❖ **70% is excreted in the feces and about 10% in the urine (as a glucuronide conjugate)**

❖ **Drug interaction**

- Bile acid sequestrants + ezetimibe....**should not be given**
- ✓ absorption of ezetimibe inhibited

Therapeutic Use

- ❖ ezetimibe is available as a 10-mg tablet that may be taken at any time during the day, with or without food
- ❖ Can be taken with other lipid lowering agents, Particularly with statins
- ❖ Statin + ezetimibe
 - ✓ prevents both the **enhanced cholesterol** synthesis induced by **ezetimibe** and the increase in **cholesterol absorption** induced by statins,
 - ✓ provide additive reductions in LDL-C levels.
 - ✓ A combination tablet containing ezetimibe, 10 mg, and various doses of simvastatin (10, 20, 40, and 80 mg) has been approved.

- Other than **rare allergic reactions**, specific adverse effects have not
 - ❖ contraindicated in pregnant and nursing women,
 - ❖ combination products(ezetimibe and a statin) should not be used by women in childbearing years in the absence of contraception.

Bile Acid-Binding Resins (RESINS)

- Colestipol
- Cholestyramine
- Colesevelam

Bile Acid-Binding Resins

- ❖ The bile acid sequestrants **cholestyramine** and **colestipol** are among the oldest of the **hypolipidemic drugs** and are probably **the safest**
- ✓ they are not absorbed from the intestine.
- ❖ These resins also are recommended for patients 11–20 years of age.
- ❖ The resins could be utilized as a second agent if statin therapy does not lower LDL-C levels sufficiently or **in cases of statin intolerance**

Mechanism of Action

- ❖ The bile acid sequestrants are **highly positively charged** and **bind negatively charged bile acids**.
- ❖ Because of their large size, the resins **are not absorbed**, and **the bound bile acids are excreted in the stool**.
- ❖ Because more than 95% of bile acids are normally reabsorbed,
- ❖ interruption of this process **depletes the pool of bile acids**, and hepatic **bile acid synthesis increases**.
 - ✓ As a result, hepatic cholesterol content declines (**conversion (Cholesterol.....Bile acid)**)
- ❖ stimulating the production of LDL receptors, an effect similar to that of statins
 - ✓ but this effect is partially offset by the enhanced cholesterol synthesis caused by upregulation of HMG-CoA reductase

Mechanism of Action...

❖ The resin-induced increase in **bile acid production** is accompanied by an increase in **hepatic triglyceride synthesis**, which is of consequence in patients with significant hypertriglyceridemia (baseline triglyceride level > 250 mg/dL).

- Use of **colesevelam** to lower LDL-C levels in hypertriglyceridemic patients should be accompanied by frequent (every 1–2 weeks) monitoring of fasting triglyceride levels

Effects on Lipoprotein Levels.

- ❖ The reduction in LDL-C by resins is dose dependent.
- ✓ Doses of **8–12 g of cholestyramine** or **10–15 g** of colestipol
 - 12%–18% reductions in LDL-C.
- ✓ Colesevelam
 - lowers LDL-C by 18% at its maximum dose.
- ✓ Maximal doses (24 g of cholestyramine, 30 g of colestipol)
 - ✓ may reduce LDL-C by as much as 25%
 - but will cause GI side effects, which are often unacceptable.
- ❖ One to 2 weeks is sufficient to attain maximal **LDL-C reduction** by a given resin dose.
- ❖ In patients with **normal TG****TG may (+)** increase transiently and then **return to baseline**.

Preparations and Use.

- ❖ **Usually Powdered forms.....cholestyramine (4 g/ dose) and colestipol (5 g/dose tablet form is also available)**
- ❖ **Preparation.....mixed with a fluid (water or juice) and drunk as a slurry or mixed with crushed ice in a blender**
 - ✓ **should take the resins before breakfast and before supper**
 - ✓ **Colesevelam hydrochloride is available as a solid tablet containing 0.625 g of colesevelam and as a powder in packets of 3.75 g or 1.875 g.**

❖ hyperchloremic acidosis....**Rare Case**

➤ Because they are administered as chloride salts

❖ Drinking a slurry of **powdered cholestyramine** or **colestipol** produces

✓ a gritty sensation that is unpleasant but generally tolerated

✓ Colesevelam is available as a hard capsule....minimize GI irritation

• **Colesevelam** << **than Colestipol** to cause the dyspepsia, bloating, and constipation.

❖ **Constipation** may occur but

✓ be prevented by adequate daily water intake and psyllium

Drug Interaction

❖ **Cholestyramine** and **colestipol** bind and interfere with the absorption of many drugs, including

- ✓ thiazides, Furosemide , propranolol,

- ✓ l-thyroxine, digoxin, warfarin, and some of the statins

- ✓ Colesevelam does not appear to interfere with the absorption

- ✓ The safety and efficacy of **colesevelam** have **not been studied in pediatric** patients or **pregnant women**